20. ROS 2 Recording and Playback Tool

1. Introduction to the Recording and Playback Tool

The commonly used recording and playback tool in ROS 2—bag2—is used to record topic data. We can use this command to store topic data as a file; afterwards, without starting a node, we can directly publish the topic data in the bag file.

This tool is very useful when we build a real robot. For example, we can record a segment of topic data when a problem occurs on the robot. After recording, we can publish it multiple times for testing and experimentation, and we can also share the topic data with others to verify algorithms.

Let us try using the bag tool to record topic data and replay it a second time.

2. Usage Tutorial

2.1. Start the Topic Node to Be Recorded

For example, the talker in the ros2 demo:

```
ros2 run demo_nodes_py talker
```

2.2. Recording

`/topic-name` is the topic name

```
# Record a single topic
ros2 bag record /topic-name
# Record multiple topics
ros2 bag record topic-name1 topic-name2
# Record all topics
ros2 bag record -a
```

Other options

-o name Customize the name of the output file

```
ros2 bag record -o file-name topic-name
```

-s Storage format

ROS 2 Jazzy uses the MCAP storage format by default. You can also use `-s` to specify an installed storage plugin, for example `sqlite3`:

```
ros2 bag record -s sqlite3 /topic-name
```

2.3. Viewing the Information of the Recorded Topics

Before playing a video, we can check the relevant information of the video through the file information, such as the recording time, size, type, and number of the topics

```
# Assume the recorded file is rosbag2_2023_10_31-07_58_23
ros2 bag info rosbag2_2023_10_31-07_58_23
```

2.4. Playing and Viewing

2.4.1. Playing

Next we can replay the data. Use the following command to play the data

```
ros2 bag play rosbag2_2023_10_31-07_58_23
```

2.4.2. Viewing

Use the ros2 topic command to view the data

```
ros2 topic echo /chatter
```

2.4.3. Playback Options

1. Playback at Multiples of Speed -r

The -r option can modify the playback rate, for example -r value; for example, -r 10 means 10x speed, playing the topic at ten times the speed

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -r 10
```

2. Loop Playback -l

This is the single-track loop

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -l
```

3. Play a Single Topic

```
ros2 bag play rosbag2_2023_10_31-07_58_23 --topics /chatter
```

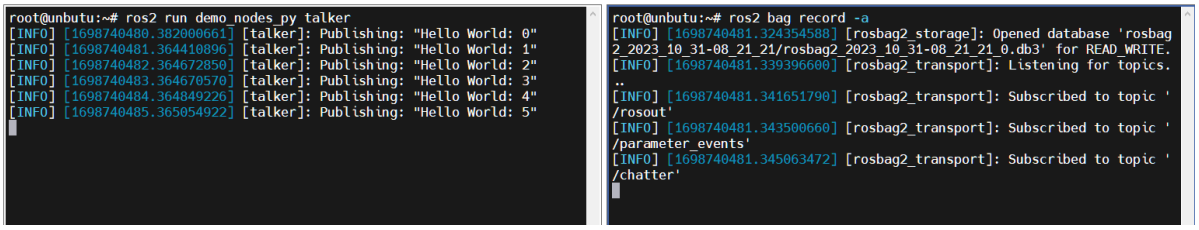
3. Example

3.1. Running the talker Node

```
ros2 run demo_nodes_py talker
```

3.2. Recording

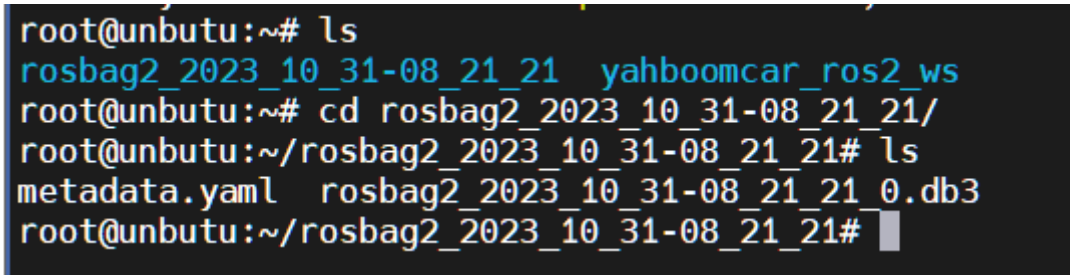
```
# Record all topics
ros2 bag record -a
```



How do you stop recording? Just use the `Ctrl+C` command in the terminal to interrupt the recording

Then you will find an extra folder in the terminal, named `rosbag2_2023_10_31-08_21_21`

Open the folder and you can see the contents



This completes the recording.

3.3. Playing and Viewing

Here we play in a loop

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -l
```

Open another terminal to view the topic:

```
ros2 topic echo /chatter
```

